Experiment No-23 : Family Tree

Aim:

To write a Prolog program to implement a family tree and find the relationships between its members.

Programming Code:

```
% ---------- Facts ----------
male(john).   male(david).  male(mike).
male(tom).    male(sam).
female(mary). female(lisa). female(anna).
female(sara). female(emma).

% parent(Parent, Child)
parent(john, david).   parent(mary, david).
parent(john, lisa).    parent(mary, lisa).
parent(david, tom).    parent(anna, tom).
parent(david, sara).   parent(anna, sara).
parent(mike, sam).     parent(lisa, sam).
parent(mike, emma).    parent(lisa, emma).

% ---------- Rules ----------
father(F, C)  :- parent(F, C), male(F).
mother(M, C)  :- parent(M, C), female(M).

son(S, P)      :- parent(P, S), male(S).
daughter(D, P) :- parent(P, D), female(D).

sibling(X, Y)  :- father(F, X), father(F, Y), X \= Y.
brother(B, X)  :- male(B), sibling(B, X).
sister(S, X)   :- female(S), sibling(S, X).

grandparent(G, C) :- parent(G, P), parent(P, C).
grandfather(G, C) :- grandparent(G, C), male(G).
grandmother(G, C) :- grandparent(G, C), female(G).

uncle(U, N) :- parent(P, N), brother(U, P).
aunt(A, N)  :- parent(P, N), sister(A, P).

cousin(X, Y) :- parent(P1, X), parent(P2, Y), sibling(P1, P2).
```

OUTPUT:

```
$ swipl -q exp23.pl

?- father(john, X).
X = david ;
X = lisa.

?- mother(M, tom).
M = anna.

?- sibling(tom, X).
X = sara.

?- grandparent(G, tom).
G = john ;
G = mary.

?- grandmother(G, sam).
G = mary.

?- uncle(U, sam).
U = david.

?- aunt(A, tom).
A = lisa.

?- cousin(tom, X).
X = sam ;
X = emma.

?- sister(emma, sam).
true.

$
```

Result

The family tree was implemented successfully using Prolog. Using the parent, male and female facts, the program derives relationships such as father, mother, sibling, grandparent, uncle, aunt and cousin.